

Table : Area under standard normal curve from 0 to $\frac{x - \mu}{\sigma}$

$\frac{x - \mu}{\sigma}$	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	.0000	.0040	.0080	.0120	.0159	.0199	.0239	.0279	.0319	.0359
0.1	.0398	.0438	.0478	.0517	.0557	.0596	.0636	.0675	.0714	.0753
0.2	.0793	.0832	.0871	.0910	.0948	.0987	.1026	.1084	.1103	.1141
0.3	.1179	.1217	.1255	.1293	.1331	.1368	.1406	.1443	.1480	.1517
0.4	.1554	.1591	.1628	.1664	.1700	.1736	.1772	.1808	.1844	.1879
0.5	.1916	.1950	.1985	.2019	.2054	.2088	.2123	.2157	.2190	.2224
0.6	.2258	.2291	.2324	.2357	.2389	.2422	.2454	.2486	.2518	.2549
0.7	.2580	.2611	.2642	.2671	.2704	.2734	.2764	.2794	.2823	.2852
0.8	.2881	.2910	.2939	.2967	.2995	.3023	.3051	.3078	.3106	.3133
0.9	.3159	.3186	.3212	.3238	.3264	.3289	.3315	.3340	.3365	.3389
1.0	.3413	.3438	.3461	.3485	.3508	.3531	.3554	.3577	.3599	.3621
1.1	.3643	.3665	.3686	.3708	.3729	.3749	.3770	.3790	.3810	.3830
1.2	.3849	.3869	.3888	.3907	.3925	.3944	.3962	.3980	.3997	.4015
1.3	.4032	.4049	.4066	.4082	.4099	.4115	.4131	.4147	.4162	.4177
1.4	.4192	.4207	.4222	.4236	.4251	.4265	.4279	.4292	.4306	.4319
1.5	.4332	.4345	.4357	.4370	.4382	.4394	.4406	.4418	.4430	.4441
1.6	.4452	.4463	.4474	.4485	.4495	.4505	.4515	.4525	.4535	.4545
1.7	.4564	.4564	.4573	.4582	.4591	.4599	.4608	.4616	.4625	.4633
1.8	.4641	.4649	.4656	.4664	.4671	.4678	.4686	.4693	.4699	.4706
1.9	.4713	.4719	.4726	.4732	.4738	.4744	.4750	.4756	.4762	.4767
2.0	.4772	.4778	.4783	.4788	.4793	.4798	.4803	.4808	.4812	.4817
2.1	.4821	.4826	.4830	.4834	.4838	.4842	.4846	.4850	.4854	.4857
2.2	.4861	.4865	.4868	.4871	.4875	.4878	.4881	.4884	.4887	.4892
2.3	.4893	.4896	.4898	.4901	.4904	.4906	.4909	.4911	.4913	.4916
2.4	.4918	.4920	.4922	.4925	.4927	.4929	.4931	.4932	.4934	.4936
2.5	.4938	.4940	.4941	.4943	.4945	.4946	.4948	.4949	.4951	.4952
2.6	.4953	.4955	.4956	.4957	.4959	.4960	.4961	.4962	.4963	.4964
2.7	.4965	.4966	.4967	.4968	.4969	.4970	.4971	.4972	.4973	.4974
2.8	.4974	.4975	.4976	.4977	.4977	.4978	.4979	.4980	.4980	.4981
2.9	.4981	.4982	.4983	.4983	.4984	.4984	.4985	.4985	.4986	.4986
3.0	.4987	.4987	.4987	.4988	.4988	.4989	.4989	.4989	.4990	.4990
3.1	.4990	.4991	.4991	.4991	.4992	.4992	.4992	.4992	.4993	.4993

SOLVED EXAMPLES

Example 1 : For a normally distributed variate with mean 1 and standard deviation 3, find the probabilities that

(i) $3.43 \leq x \leq 6.19$ (ii) $-1.43 \leq x \leq 6.19$

Solution : Given $\mu = 1$ and $\sigma = 3$

(i) When $x = 3.43$, $z = \frac{x - \mu}{\sigma} = \frac{3.43 - 1}{3} = \frac{2.43}{3} = 0.81 = z_1$ (say)

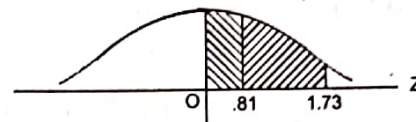
When $x = 6.19$, $z = \frac{x - \mu}{\sigma} = \frac{6.19 - 1}{3} = \frac{5.19}{3} = 1.73 = z_2$ (say)

$\therefore P(3.43 \leq x \leq 6.19) = P(0.81 \leq z \leq 1.73)$

$= |A(z_2) - A(z_1)|$

$= |A(1.73) - A(0.81)| = 0.4582 - 0.2910$ (from tables)

$= 0.1672$ (Cross hatched area in the figure)



(ii) When $x = -1.43$, $z = \frac{x - \mu}{\sigma} = \frac{-1.43 - 1}{3} = -0.81 = z_1$ (say)

When $x = 6.19$, $z = \frac{x - \mu}{\sigma} = \frac{6.19 - 1}{3} = 1.73 = z_2$ (say)

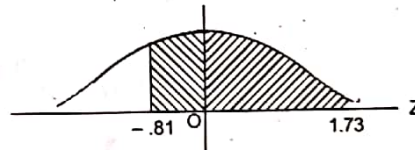
$\therefore P(-1.43 \leq x \leq 6.19) = P(-0.81 \leq z \leq 1.73)$

$= A(z_2) + A(z_1)$

$= A(1.73) + A(-0.81)$

$= A(1.73) + A(0.81)$ [$\because A(-z) = A(z)$]

$= 0.4582 + 0.2910 = 0.7492 =$ shaded area in the figure.



Example 2 : If X is a normal variate with mean 30 and standard deviation 5.

Find the probabilities that (i) $26 \leq X \leq 40$ (ii) $X \geq 45$.

[JNTU 2000, 2000S, JNTU (K) Nov. 2009 (Set No. 1)]

Solution : Given mean, $\mu = 30$ and S.D., $\sigma = 5$

(i) When $x = 26$, $z = \frac{x - \mu}{\sigma} = \frac{26 - 30}{5} = -0.8 = z_1$ (say)

When $x = 40$, $z = \frac{x - \mu}{\sigma} = \frac{40 - 30}{5} = 2 = z_2$ (say)

$\therefore P(26 \leq X \leq 40) = P(-0.8 \leq z \leq 2)$

$= A(z_2) + A(z_1) = A(2) + A(-0.8)$

$= 0.4772 + 0.2881 = 0.7653$ (from normal distribution tables)

* See Normal Distribution Table given, read the data in row 1.7 and column 0.03.

(ii) When $x = 45$, $z = \frac{x - \mu}{\sigma} = \frac{45 - 30}{5} = 3 = z_1$ (say)

$$\begin{aligned}\therefore P(X \geq 45) &= P(z_1 \geq 3) \\ &= 0.5 - A(z_1) = 0.5 - A(3) \\ &= 0.5 - 0.49865 = 0.00135\end{aligned}$$

✓ **Example 3 :** In a Normal distribution, 7% of the items are under 35 and 89% are under 63. Determine the mean and variance of the distribution.

[JNTU Jan. 2007, (H) Dec. 2011, (K) Nov. 2012 (Set No. 2)]

(OR) Find the mean and standard deviation of a normal distribution in which 7% of items are under 35 and 89% are under 63. [JNTU (H) Nov., 2009, (A) Nov. 2010 (Set No. 4)]

Solution : Let μ be the mean (at $z = 0$) and σ the standard deviation of the normal curve. 7% of the items are under 35 means the area to the left of the ordinate $x = 35$.

Given $P(X < 35) = 0.07$ and $P(X < 63) = 0.89$

$$\therefore P(X > 63) = 1 - P(X < 63) = 1 - 0.89 = 0.11$$

The points $X = 35$ and $X = 63$ are shown in the following Figure.

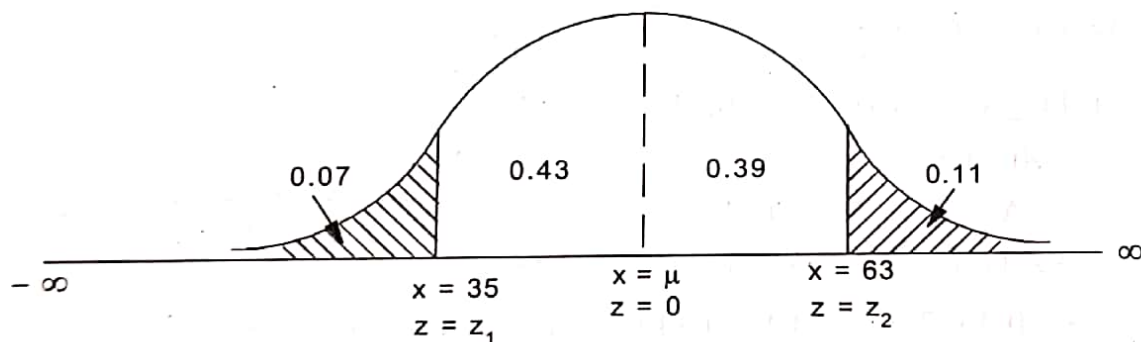


Fig.

When $x = 35$, $z = \frac{x - \mu}{\sigma} = \frac{35 - \mu}{\sigma} = -z_1$ (say) ... (1)

When $x = 63$, $z = \frac{x - \mu}{\sigma} = \frac{63 - \mu}{\sigma} = z_2$ (say) ... (2)

From the figure, we have

$$P(0 < z < z_2) = 0.39 \Rightarrow z_2 = 1.23 \quad (\text{from tables})$$

and $P(0 < z < z_1) = 0.43 \Rightarrow z_1 = 1.48$

From (1), we have $\frac{35 - \mu}{\sigma} = -1.48$... (3)

From (2), we have $\frac{63 - \mu}{\sigma} = 1.23$... (4)

(4) - (3) gives

$$\frac{28}{\sigma} = 2.71 \Rightarrow \sigma = \frac{28}{2.71} = 10.332$$

From (3), $35 - \mu = -1.48 (\sigma) = (-1.48)(10.332) = -15.3$

$$\therefore \mu = 35 + 15.3 = 50.3$$

and variance $= \sigma^2 = 106.75$

- Example 4 :** In a normal distribution 31% of the items are under 45 and 8% are over 64. Find the mean and variance of the distribution.

[JNTU (A) Dec. 2009, (K) Nov. 2012, Mar. 2014 (Set No. 2)]

Solution : Let X be the continuous random variable.

Let μ be the mean and σ the standard deviation.

Given $P(X < 45) = 0.31$ and $P(X > 64) = 0.08$

Standard variable, $z = \frac{X - \mu}{\sigma}$

When $X = 45$, let $z = z_1$

$$\text{so that } z_1 = \frac{45 - \mu}{\sigma} \quad \dots (1)$$

$$\therefore \int_{-\infty}^{z_1} \phi(z) dz = 0.31 \quad \text{or} \quad \int_{-\infty}^0 \phi(z) dz - \int_{z_1}^0 \phi(z) dz = 0.31$$

$$\therefore \int_{z_1}^0 \phi(z) dz = \int_{-\infty}^0 \phi(z) dz - 0.31 = 0.5 - 0.31 = 0.19$$

$$\text{Hence } P(0 < z < z_1) = 0.19 \Rightarrow z_1 = -0.5 \text{ (from table)} \quad \dots (2)$$

$$\text{When } X = 64, z = \frac{64 - \mu}{\sigma} = z_2 \text{ (say)} \quad \dots (3)$$

$$\therefore \int_{z_2}^{\infty} \phi(z) dz = 0.08 \quad \text{or} \quad \int_0^{\infty} \phi(z) dz - \int_0^{z_2} \phi(z) dz = 0.08$$

$$\text{Hence } \int_0^{z_2} \phi(z) dz = \int_0^{\infty} \phi(z) dz - 0.08 = 0.5 - 0.08 = 0.42$$

$$\text{Thus } P(0 < z < z_2) = 0.42 \Rightarrow z_2 = 1.4 \text{ (from tables)} \quad \dots (4)$$

$$\text{From (1) and (2), we have } \frac{45 - \mu}{\sigma} = -0.5 \Rightarrow 45 - \mu = -0.5\sigma \quad \dots (5)$$

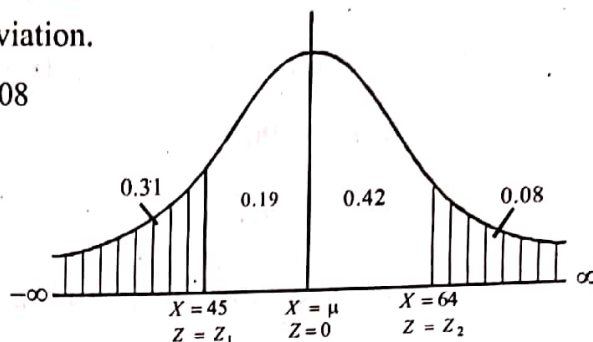
$$\text{From (3) and (4), we have } \frac{64 - \mu}{\sigma} = 1.4 \Rightarrow 64 - \mu = 1.4\sigma \quad \dots (6)$$

$$(5) - (6) \text{ gives } (45 - \mu) - (64 - \mu) = -0.5\sigma - 1.4\sigma$$

$$\Rightarrow -19 = -1.9\sigma \quad \therefore \sigma = \frac{19}{1.9} = 10$$

$$\text{From (5), } \mu = 45 + 0.5\sigma = 45 + 0.5(10) = 50$$

Hence mean = 50 and standard deviation = 10



Example 5 : Suppose 10 percent of the probability for a normal distribution $n(\mu, \sigma^2)$ is below 35 and 5 percent above 90. What are the values of μ and σ ? [JNTU (K) 2009 (Set No.2)]

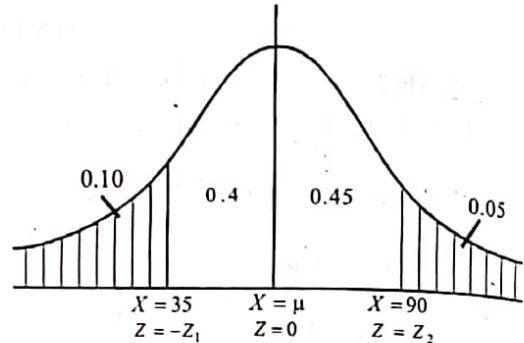
Solution : If $X \sim N(\mu, \sigma^2)$ then we are given $P(X < 35) = 0.10$ and $P(X > 90) = 0.05$. The points $X = 35$ and $X = 90$ are located as shown in Figure.

$$\text{Standard variable, } z = \frac{X - \mu}{\sigma}$$

$$\text{When } X = 35, z = \frac{35 - \mu}{\sigma} = -z_1 \text{ (say)}$$

and when $X = 90$,

$$z = \frac{90 - \mu}{\sigma} = z_2 \text{ (say)}$$



Thus we have $P(0 < z < z_1) = 0.4$ and $P(0 < z < z_2) = 0.45$.

Hence from tables, we have $z_1 = 1.3$ and $z_2 = 1.65$

$$\therefore \frac{35 - \mu}{\sigma} = -1.3 \text{ and } \frac{90 - \mu}{\sigma} = 1.65$$

$$\text{Subtracting, we get } \frac{55}{\sigma} = 2.95 \Rightarrow \sigma = \frac{55}{2.95} = 18.64$$

$$\therefore \mu = 35 + 1.3\sigma = 35 + 1.3(18.64) = 35 + 24.24 = 59.24$$

Hence $\mu = 59.24$ and $\sigma = 18.64$

Example 6 : The mean and standard deviation of the marks obtained by 1000 students in an examination are respectively 34.5 and 16.5. Assuming the normality of the distribution, find the approximate number of students expected to obtain marks between 30 and 60.

[JNTU (A) Nov. 2010 (Set No.1)]

Solution : Given mean, $\mu = 34.5$ and S.D., $\sigma = 16.5$

$$\text{When } x = 30, z = \frac{x - \mu}{\sigma} = \frac{30 - 34.5}{16.5} = -0.27 = z_1 \text{ (say)}$$

$$\text{When } x = 60, z = \frac{x - \mu}{\sigma} = \frac{60 - 34.5}{16.5} = 1.54 = z_2 \text{ (say)}$$

$$\begin{aligned} \therefore P(30 \leq x \leq 60) &= P(z_1 \leq z \leq z_2) \\ &= A(z_2) + A(z_1) \quad [\because z_1 < 0 \text{ and } z_2 > 0] \\ &= A(1.54) + A(-0.27) = A(1.54) + A(0.27) \\ &= 0.4382 + 0.1084 \quad [\text{From normal distribution tables}] \\ &= 0.5916 \end{aligned}$$

$$\begin{aligned} \therefore \text{The number of students who get marks between 30 and 60} \\ &= 0.5916 \times 1000 = 591.6 \end{aligned}$$

Hence 592 students get marks between 30 and 60.

Example 7 : The marks obtained in mathematics by 1000 students is normally distributed with mean 78% and standard deviation 11%. Determine

- How many students got marks above 90%
- What was the highest mark obtained by the lowest 10% of the students
- Within what limits did the middle of 90% of the students lie.

[JNTU 2006S, 2007, (A) Nov. 2010, Apr. 2012, (K) Nov. 2011 (Set No. 2)]

Solution : Given mean, $\mu = 0.78$ and S.D., $\sigma = 0.11$

(i) When $x = 0.9$, $z = \frac{x - \mu}{\sigma} = \frac{0.9 - 0.78}{0.11} = 1.09 = z_1$ (say)

Hence the number of students with marks more than 90%
 $= 0.1379 \times 1000$
 $= 137.9 \approx 138$

- (ii) The 0.1 area to the left of z corresponds to the lowest 10% of the students

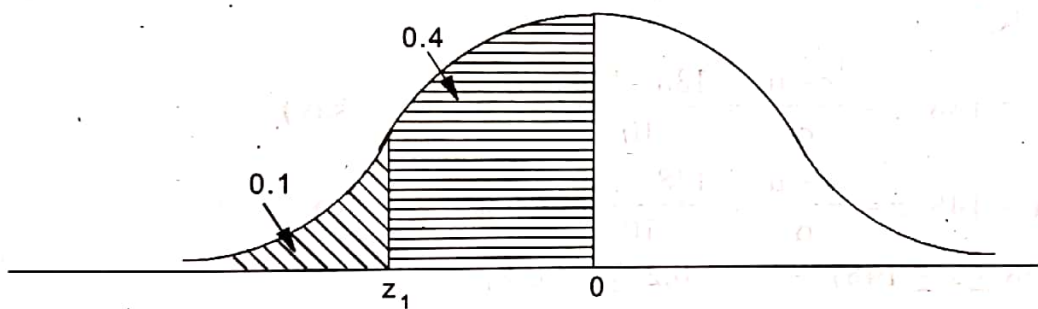


Fig.

From Figure,

$$0.4 = 0.5 - 0.1 = 0.5 - \text{Area from } 0 \text{ to } z_1$$

$$\therefore z_1 = -1.28 \text{ (from tables)}$$

$$\text{Thus } -1.28 = \frac{x - \mu}{\sigma} = \frac{x - 0.78}{0.11} \Rightarrow x = 0.78 - 1.28(0.11) = 0.6392$$

Hence the highest mark obtained by the lowest 10% of students
 $= 0.6392 \times 1000 \approx 64\%$

- (iii)

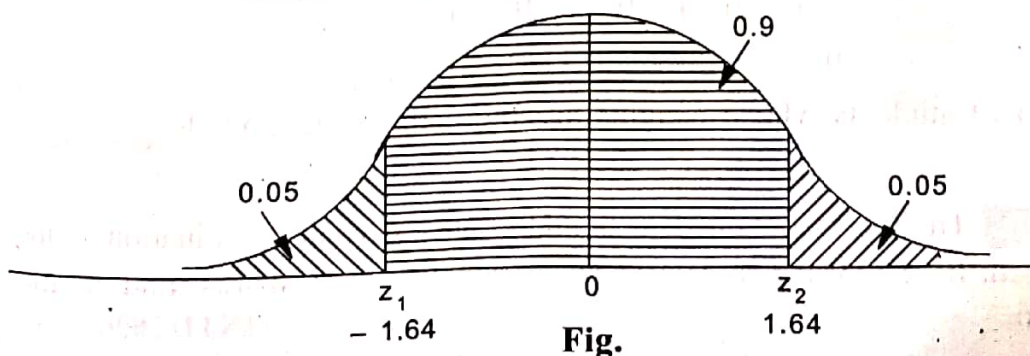


Fig.

Middle 90% correspond to 0.9 area, leaving 0.05 area on both sides. Then the corresponding z 's are ± 1.64 [since if the area from 0 to z is 0.45, then $z = 1.64$]

$$\therefore -1.64 = z_1 = \frac{x_1 - \mu}{\sigma} = \frac{x_1 - 0.78}{0.11}$$

$$\Rightarrow x_1 = 0.78 - 1.64(0.11) = 0.5996 \text{ or } 59.96\%$$

$$\text{and } 1.64 = z_2 = \frac{x_2 - \mu}{\sigma} = \frac{x_2 - 0.78}{0.11}$$

$$\Rightarrow x_2 = 1.64(0.11) + 0.78 = 0.9604 \text{ or } 96.04\%$$

Thus the middle 90% have marks in between 60 to 96.

Example 8 : Suppose the weights of 800 male students are normally distributed with mean $\mu = 140$ pounds and standard deviation 10 pounds. Find the number of students whose weights are (i) between 138 and 148 pounds (ii) more than 152 pounds

[JNTU 2006 S (Set No.2)]

Solution : Let μ be the mean and σ be the standard deviation. Then $\mu = 140$ pounds and $\sigma = 10$ pounds.

$$(i) \text{ When } x = 138, z = \frac{x - \mu}{\sigma} = \frac{138 - 140}{10} = -0.2 = z_1 \text{ (say)}$$

$$\text{When } x = 148, z = \frac{x - \mu}{\sigma} = \frac{148 - 140}{10} = 0.8 = z_2 \text{ (say)}$$

$$\begin{aligned} \therefore P(138 \leq x \leq 148) &= P(-0.2 \leq z \leq 0.8) \\ &= A(z_2) + A(z_1) = A(0.8) + A(-0.2) \\ &= A(0.8) + A(0.2) \\ &= 0.2881 + 0.0793 = 0.3674 \end{aligned}$$

Hence the number of students whose weights are between 138 pounds and 148 pounds $= 0.3674 \times 800 \approx 294$.

$$(ii) \text{ When } x = 152, z = \frac{x - \mu}{\sigma} = \frac{152 - 140}{10} = 1.2 = z_1 \text{ (say)}$$

$$\begin{aligned} \therefore P(x > 152) &= P(z > z_1) \\ &= 0.5 - A(z_1) = 0.5 - A(1.2) \\ &= 0.5 - 0.3849 = 0.1151 \end{aligned}$$

$$\begin{aligned} \therefore \text{Number of students whose weights are more than 152 pounds} \\ &= 800 \times 0.1151 = 92 \end{aligned}$$

Example 9 : The

Example 14 : In an examination it is laid down that a student passes if he secures 40 percent or more. He is placed in the first, second and third division according as he secures 60% or more marks, between 50% and 60% marks and marks between 40% and 50% respectively. He gets a distinction in case he secures 75% or more. It is noticed from the results that 10% of the students failed in the examination, whereas 5% of them obtained distinction. Calculate the percentage of students placed in the second division. (Assume normal distribution of marks).

[JNTU (K) May 2013 (Set No. 1)]

Solution : Let μ be the mean and σ the standard deviation of the normal distribution of marks.

Let the variable X denote the marks in the examination. Then we are given

$$P(X < 40) = 0.10 \text{ and } P(X \geq 75) = 0.05$$

$$\text{When } X = 40, z = \frac{X - \mu}{\sigma} = \frac{40 - \mu}{\sigma} = -z_1 \text{ (say)}$$

$$\text{When } X = 75, z = \frac{X - \mu}{\sigma} = \frac{75 - \mu}{\sigma} = z_2 \text{ (say)}$$

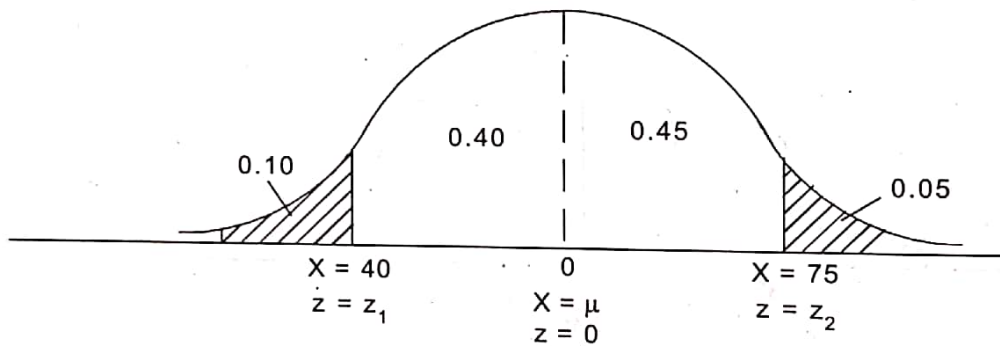


Fig.

$$\therefore P(0 < z < z_2) = 0.5 - 0.05 = 0.45$$

$$\text{and } P(0 < z < z_1) = P(-z_1 < z < 0) = 0.5 - 0.10 = 0.40$$

$$\therefore \text{From normal tables, we have } z_1 = 1.28 \text{ and } z_2 = 1.64$$

$$\text{Hence } \frac{40 - \mu}{\sigma} = -1.28 \Rightarrow \frac{\mu - 40}{\sigma} = 1.28 \dots (1) \text{ and } \frac{75 - \mu}{\sigma} = 1.64 \dots (2)$$

$$(1) + (2) \text{ gives } \frac{35}{\sigma} = 2.92 \Rightarrow \sigma = \frac{35}{2.92} = 11.99 \approx 12$$

$$\text{From (1), } \mu = 40 + \sigma(1.28) = 40 + 12(1.28) \approx 55$$

The probability 'p' that a candidate is placed in the second division is equal to the probability that his score lies between 50 and 60. That is,

$$\begin{aligned} p &= P(50 < X < 60) = P(-0.42 < Z < 0.42) \quad \left[\because Z = \frac{X - 55}{12} \right] \\ &= A(0.42) + A(-0.42) \\ &= A(0.42) + A(0.42) \quad (\text{due to symmetry}) \\ &= 2A(0.42) = 2(0.1628) = 0.3256 = 0.32 \text{ (approximately)} \end{aligned}$$

Hence 32% candidates get second division in the examination.

Example 15 : A manufacturer knows from experience that the resistance of resistors he produces is normal with mean 100 ohms and standard deviation 2 ohms. What percentage of resistors will have resistance between 98 ohms and 102 ohms ?

Solution : Let μ be the mean and σ the standard deviation of the normal distribution. Then we are given that

$$\mu = 100 \text{ ohms and } \sigma = 2 \text{ ohms}$$

Let the variable X represents the resistance of resistors

$$\text{When } X = 98 \text{ ohms, } z = \frac{X - \mu}{\sigma} = \frac{98 - 100}{2} = -1$$

$$\text{and when } X = 102 \text{ ohms, } z = \frac{X - \mu}{\sigma} = \frac{102 - 100}{2} = 1$$

$$\begin{aligned} \therefore P(98 < X < 102) &= P(-1 < z < 1) \\ &= A(1) + A(-1) = A(1) + A(1) = 2A(1) \\ &= 2(0.3413) = 0.6826 \end{aligned}$$

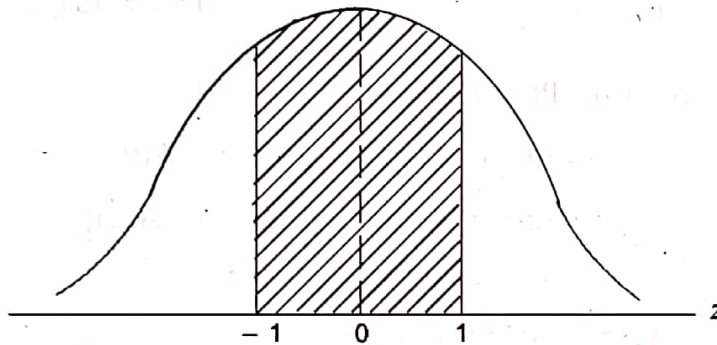


Fig.

$\therefore$ Percentage of resistors having resistance between 98 ohms and 102 ohms
 $= 68.26 = 68$ (approximately).

Example 16 : Given that the mean height of students in a class is 158 cms with standard deviation of 20 cms. Find how many students heights lie between 150 cms and 170 cms, if there are 100 students in the class. [JNTU 2007, 2007S, (H) Dec. 2011 (Set No. 4)]

Solution : We have

Mean, $\mu = 158$ cm and standard deviation, $\sigma = 20$ cms

$$\therefore z = \frac{X - \mu}{\sigma} = \frac{X - 158}{20}$$

$$\text{When } X = 150, z = \frac{150 - 158}{20} = \frac{-8}{20} = -0.4$$

$$\text{When } X = 170, z = \frac{170 - 158}{20} = \frac{12}{20} = 0.6$$

$$\begin{aligned} \therefore P(150 \leq X \leq 170) &= P(-0.4 \leq z \leq 0.6) \\ &= P(-0.4 \leq z \leq 0) + P(0 \leq z \leq 0.6) \\ &= P(0 \leq z \leq 0.4) + P(0 \leq z \leq 0.6), \text{ due to symmetry} \\ &= 0.1554 + 0.2257 = 0.3811 \end{aligned}$$

Number of students whose height lie between 150 cms and 170 cms

$$= \text{Probability} \times \text{total no. of students}$$

$$= 0.3811 \times 100 = 38 \quad (\because \text{Number of students should be integer.})$$

Example 17 : Suppose 2% of the people on the average are left handed. Find

(i) The probability of finding 3 or more left handed

(ii) The probability of finding ≤ 1 left handed

(b) The mean and standard deviation of a normal variable are 8 and 4 respectively.

Find (i) $p(5 \leq X \leq 10)$ (ii) $p(x \geq 5)$

[JNTU (H) 2009 (Set No. 1)]

Solution : Given mean, $\mu = 8$ and standard deviation, $\sigma = 4$.

$$(i) \quad \text{When } x = 5, \quad z = \frac{x - \mu}{\sigma} = \frac{5 - 8}{4} = \frac{-3}{4} = -0.75 = z_1 \quad (\text{say})$$

$$\text{When } x = 10, \quad z = \frac{x - \mu}{\sigma} = \frac{10 - 8}{4} = \frac{2}{4} = \frac{1}{2} = 0.5 = z_2 \quad (\text{say})$$

$$\therefore P(5 \leq X \leq 10) = P(-0.75 \leq z \leq 0.5)$$

$$= A(z_2) + A(z_1) = A(-0.75) + A(0.5)$$

$$= A(0.75) + A(0.5) \quad [\because A(-z) = A(z)]$$

$$= 0.2734 + 0.1916 = 0.465$$

(ii) When $x = 5$, $z_1 = -0.75$, from above

$$\therefore P(X \geq 5) = P(z_1 \geq -0.75) = 0.5 - A(z_1)$$

$$= 0.5 - A(-0.75) = 0.5 - 0.2734 = 0.2266$$

Example 18 : In a test on 2000 electrical bulbs, it was found that the life of a particular make, was normally distributed with an average life of 2040 hours and S.D. of 40 hrs. Estimate the number of bulbs likely to burn for more than 2140.

Solution : Given mean, $\mu = 2040$ hrs and S.D., $\sigma = 40$ hrs

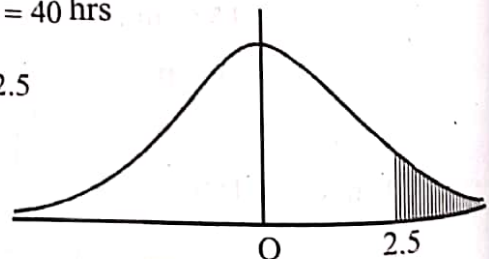
$$\text{When } x = 2140, \quad z = \frac{x - \mu}{\sigma} = \frac{2140 - 2040}{40} = \frac{100}{40} = 2.5$$

$$P(x > 2140) = P(z > 2.5) = 0.5 - P(0 \leq z \leq 2.5)$$

$$= 0.5 - 0.4938 = 0.0062$$

Hence the number of bulbs likely to burn

$$\text{for more than 2140 hours} = 0.0062 \times 2000 = 12.4 \approx 12.$$



Example 19 : If the marks obtained by a number of students in a certain subject are approximately normally distributed with mean 65 and standard deviation 5. If three students are selected at random from this group what is the probability that at least one of them would have scored above 75?

[JNTU (K) May 2010 (Set No. 2)]

Solution : Here $\mu = \text{mean} = 65$ and $\sigma = \text{S.D.} = 5$

We have $z = \frac{x - \mu}{\sigma}$ when $x = 75, z = \frac{75 - 65}{5} = \frac{10}{5} = 2$

$$\therefore p = P(x > 75) = P(z > 2) = 0.5 - P(0 \leq z \leq 2) = 0.5 - 0.4772 = 0.0228$$

Hence the required probability that out of 3 students, at least one of them will have marks over 75 is given by

$$\begin{aligned} {}^3C_1 p q^2 + {}^3C_2 p^2 q^{3-2} + {}^3C_3 p^3 q^{3-3} &= 3p(1-p)^2 + 3p^2(1-p) + p^3 \quad [\because q = 1 - p] \\ &= 3(0.0228)(1 - 0.0228)^2 + 3(0.0228)^2(1 - 0.0228) + (0.0228)^3 = 0.0668. \end{aligned}$$

Example 20 : If X is normally distributed with mean 2 and variance 0.1, then find $P(|X - 2| \geq 0.01)$?

[JNTU (K) May 2010 (Set No. 3)]

Solution : Here $\mu = 2, \sigma = 0.1$

$$\text{when } X = 1.99, z = \frac{x - \mu}{\sigma} = \frac{1.99 - 2}{0.1} = -0.1$$

$$\text{when } X = 2.01, z = \frac{2.01 - 2}{0.1} = \frac{0.01}{0.1} = 0.1$$

$$\therefore P(|X - 2| < 0.01) = P(1.99 < X < 2.01)$$

$$= P(-0.1 < z < 0.1)$$

$$= 2P(0 < z < 0.1)$$

$$= 2 \times 0.0398 = 0.0796$$

$$\therefore P(|X - 2| \geq 0.01) = 1 - P(|X - 2| < 0.01) = 1 - 0.0796 = 0.9204$$

Example 21 : It is claimed that a random sample of 100 tires with a mean life of 15269 is drawn from a population of tires which has a mean life of 15200 km and a S. D. of 1248 km. Test the claim at $\alpha = 0.05$. [JNTU (A) Nov. 2011 (Set No. 3)]

Solution : Let the Null Hypothesis is $H_0 : \mu = 15200$

Then the Alternative Hypothesis is $H_1 : \mu \neq 15200$

Since H_1 is two - tailed (i.e., two-sided), we are to use two - tailed test.

Level of significance, $\alpha = 0.05$

Let us assume that H_0 is true.

The test statistic is $z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$

Here $\bar{x}$ = sample mean = 15269, $n = 100 (> 30)$, $\mu = 15200$, $\sigma = 1248$

Since the sample size is large, $\bar{x}$ is approximately normally distributed with mean μ and

$$\text{S. D.} = \frac{\sigma}{\sqrt{n}}$$

$$\therefore z = \frac{15269 - 15200}{1248 / \sqrt{100}} = \frac{690}{1248} = 0.553.$$

The table value of z at 5% level of significance for two-tailed test is 1.96.

Since $|z| = 0.553 < 1.96$, we accept the Null hypothesis H_0 at 5% level of significance and conclude that the claim is true.

Example 22 : When the mean of marks was 50% and S. D. 5% then 60% of the students failed in an examination ? Determine the grace marks to be awarded in order to show that 70% of the students passed. Assume that the marks are normally distributed.

[JNTU (A) Apr. 2012 (Set No. 1)]

Solution : Let x be the marks obtained in the examination.

Given $\mu = 0.5$, S. D., $\sigma = 0.05$

Before grace marks were added, 60% failed. That is, 40% passed the examination.

The area $0.4 \left(< \frac{1}{2} \right)$ corresponds pass students.

Find the value of z_1 such that the area to its right is 0.4.

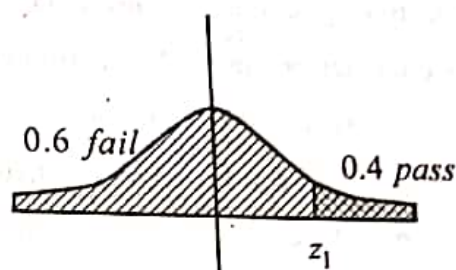
The area from 0 to z_1 is $0.5 - 0.4 = 0.1$.

From normal table for the area 0.1, the value of z is 0.25.

The standard normal variate is $z = \frac{x - \mu}{\sigma}$.

$$\Rightarrow 0.25 = z_1 = \frac{x_1 - 0.5}{0.05}$$

$$\Rightarrow x_1 = (0.25)(0.05) + 0.5 = 0.5125$$

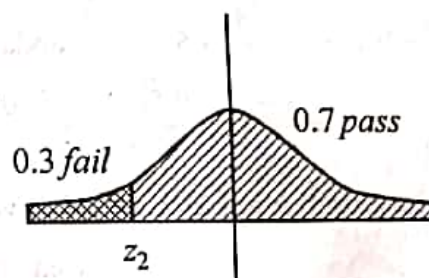


After grace marks were added, 70% passed the examination. The area $0.70 \left(> \frac{1}{2} \right)$ corresponds pass students. So determine z_2 such that the area to its right is 0.7. z_2 must be negative. The area from 0 to z_2 is $0.5 - 0.3 = 0.2$.

From table $z_2 = -0.52$.

$$\therefore z_2 = -0.52 = \frac{x_2 - 0.5}{0.05}$$

$$\Rightarrow x_2 = (-0.52)(0.05) + 0.5 = -0.026 + 0.5 = 0.474$$



$\therefore$ The minimum pass mark for a student is 51.25 before grace while the minimum pass mark is 47.40 after grace. So grace marks awarded is $51.25 - 47.40 = 3.85$

Example 23 : The mean height of students in a college is 155 cms and standard deviation 15. What is the probability that the mean height of 36 students is less than 157 cms.

[JNTU (H) Dec. 2011 (Set No. 4)]

Solution : Let X be the continuous random variable denoting the height of students. Then mean, $\mu = 155$ and S. D., $\sigma = 15$.

We know that the standardized value of X is $z = \frac{X - \mu}{\sigma}$

$$\text{when } X = 157, z = \frac{157 - 155}{15} = \frac{2}{15} = 0.13$$

Probability that the height of student is less than 157 cm.

$$= P(X < 157) = P(z < 0.13)$$

= area under the normal curve to the right of $z = 0.13$

$$= 0.0517$$

$$\therefore \text{Required probability} = 0.0517 \times 36 = 1.86$$

Example 24 : The mean inside diameter of a sample of 200 washers produced by a Machine is 500 cms with standard deviation 0.005 cms. The purpose of which these washers are intended a maximum tolerance in the diameter 0.495 to 0.505 cms. otherwise the washers are considered defective. Determine the percentage of defective washers produced by the machine, assuming the diameters are normally distributed.

[JNTU (H) Apr. 2012 (Set No. 3)]

Solution : We are given

μ = Mean of the inside diameters = 0.500 cm

σ = Standard deviation = 0.005 cm

The tolerance limits of non - defective tubes are 0.495 cm and 0.505 cm.

When $x = 0.495$ (standardized value of 0.495),

$$z = \frac{x - \mu}{\sigma} = \frac{0.495 - 0.500}{0.005} = -1$$

When $x = 0.505$ (i.e. standardized value of 0.505),

$$z = \frac{x - \mu}{\sigma} = \frac{0.505 - 0.500}{0.005} = 1$$

$\therefore$ Proportion of non-defective washers

= Area under the standard normal curve between $z = -1$ and $z = +1$

= 2 (Area between $z = 0$ and $z = 1$)

= $2 \times 0.3413 = 0.6826 = 68.26\%$

Hence the required percentage of defective washers = $100\% - 68.26\% = 31.74\%$

[(or) Probability of defective washers = $1 - \text{probability of non-defective washers}$

$$= 1 - p(0.495 \leq x \leq 0.505)$$

$$= 1 - p(-1 \leq z \leq 1)$$

$$= 1 - 2 \times p(0 \leq z \leq 1)$$

$$= 1 - 2(0.3413) = 0.3174$$

$\therefore$ Percentage of defective washers = $31.74 \approx 32\%$

Example 25 : 1000 students have written an examination the mean of test is 35 and standard deviation is 5. Assuming the distribution to be normal, find:

(i) How many students marks lie between 25 and 40?

(ii) How many students get more than 40?

(iii) How many students get below 20?

(iv) How many students get more than 50?

Solution : Here $\mu = 35$ and $\sigma = 5$

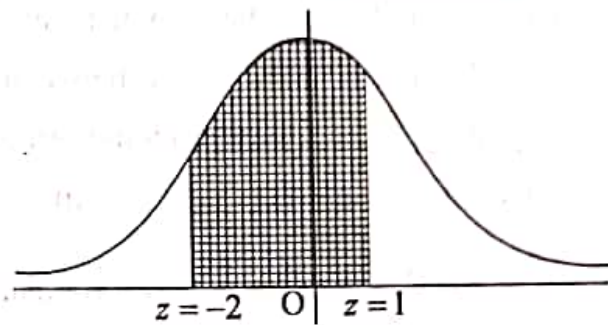
We know, standardised variate, $z = \frac{x - \mu}{\sigma} = \frac{x - 35}{5}$

When $x_1 = 25$, then

$$z = \frac{25 - 35}{5} = \frac{-10}{5} = -2$$

When $x_2 = 40$, then

$$z = \frac{40 - 35}{5} = \frac{5}{5} = 1$$



$\therefore$ Probability of students whose marks lie between 25 and 40 $= P(25 \leq X \leq 40)$

$$= P(-2 \leq z \leq 1) = P(-2 \leq z \leq 0) + P(0 \leq z \leq 1)$$

$$= P(0 \leq z \leq 2) + P(0 \leq z \leq 1), \text{ by symmetry}$$

$$= 0.4772 + 0.3415 \text{ [From Tables]}$$

$$= 0.8185$$

Number of students whose marks lie between 25 and 40 $= 1000 \times 0.8185 = 819$

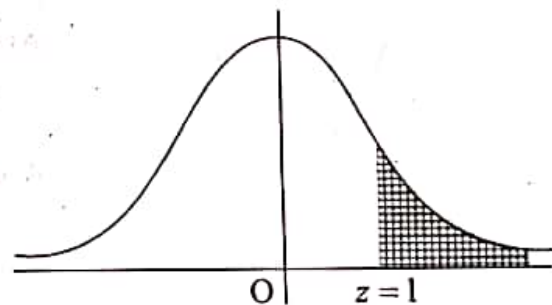
(approximately)

(ii) When $x = 40$, then $z = \frac{40 - 35}{5} = \frac{5}{5} = 1$

$$P(X > 40) = P(z > 1) = 0.5 - P(z \leq 1)$$

$$= 0.5 - 0.3415$$

$$= 0.1585 \text{ [From Tables]}$$



$\therefore$ Number of students whose marks are greater than 40 $= 1000 \times 0.1585 = 159$

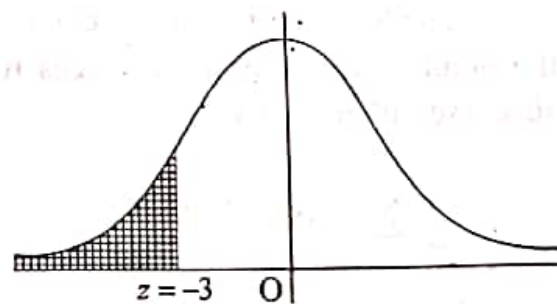
(approximately)

(iii) When $x = 20$, then $z = \frac{20 - 35}{5} = \frac{-15}{5} = -3$

$$P(X < 20) = P(z \leq -3) = 0.5 - P(-3 \leq z \leq 0)$$

$$= 0.5 - P(0 \leq z \leq 3), \text{ by symmetry}$$

$$= 0.5 - 0.499 = 0.001 \text{ [From Tables]}$$



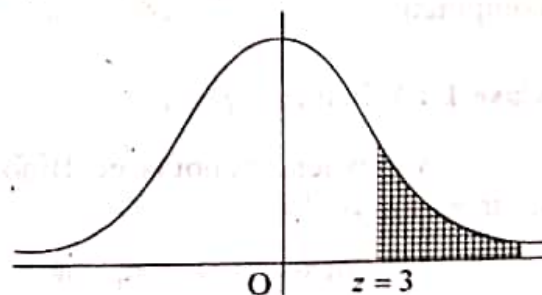
$\therefore$ Number of students whose marks are less than 20 $= 1000 \times 0.001 = 1$

(iv) When $x = 50$, $z = \frac{50 - 35}{5} = \frac{15}{5} = 3$

$$\therefore P(X > 50) = P(z > 3) = 0.5 - P(0 \leq z \leq 3)$$

$$= 0.5 - 0.499$$

$$= 0.001 \text{ [From Tables]}$$



Hence number of students whose marks are greater than 50 $= 1000 \times 0.001 = 1$.

Example 26 : An electrical firm manufactures light bulbs that have a life, before burnout, which is normally distributed with mean equal to 800h and a standard deviation of 40 h. Find the probability that a bulb burns between 778h and 834 h. [JNTU(A) Dec. 2014]

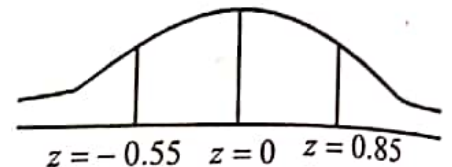
Solution : Let X be a random normal variate measuring the life time of bulbs.

Here mean $\mu = 800$ and $\sigma = 40$

$\therefore Z = \frac{X - \mu}{\sigma} = \frac{X - 800}{40}$ is a standard normal variate $N(0,1)$

When $X = 778$, then $Z = \frac{778 - 800}{40} = \frac{-22}{40} = -0.55$

When $X = 834$, then $Z = \frac{834 - 800}{40} = \frac{34}{40} = 0.85$



When X lies between 778 and 834, then Z lies between -0.55 and 0.85

$\therefore$ Required probability = Area under the curve from $z = -0.55$ to $z = 0.85$

= Area under the curve from $(z = -0.55$ to $z = 0)$

+ Area under the curve from $(z = 0$ to $z = 0.85)$

= Area under the curve from $(z = 0$ to $z = 0.55)$

+ Area under the curve from $(z = 0$ to $z = 0.85)$

(symmetric property)

$$= 0.2088 + 0.3023 = 0.5111$$

7.16 NORMAL APPROXIMATION TO THE BINOMIAL DISTRIBUTION

The Normal distribution can be used to approximate the Binomial distribution. Suppose the number of successes x ranges from x_1 to x_2 . Then the probability of getting x_1 to x_2 successes is given by

$$\sum_{r=x_1}^{x_2} nC_r p^r q^{n-r}$$

For large n , the calculation of binomial probabilities is very difficult. In such cases the binomial curve can be replaced by the normal curve and the required probability is computed. We consider two cases.

Case 1 : When $p = q = \frac{1}{2}$

Even when n is not large, Binomial distribution (B.D) can be approximated by Normal distribution (N.D).

Mean of the B.D. = np and S.D. = $\sqrt{npq}$

Hence for the corresponding Normal distribution, μ and σ are known.

We know that $z = \frac{x - \mu}{\sigma}$

Let z_1 and z_2 be the values of z , corresponding to x_1 and x_2 of x respectively. Then

$$P(x_1 < x < x_2) = P(z_1 < z < z_2) = \int_{z_1}^{z_2} \phi(z) dz$$

and this can be determined by using the normal tables.

Case 2 : When $p \neq q$.

For large n , we can approximate the binomial curve by the normal curve and calculate the probability.

For any success x , real class interval is $\left(x - \frac{1}{2}, x + \frac{1}{2}\right)$. Hence z_1 must correspond to the lower limit of the x_1 class and z_2 , to the upper limit of the x_2 class.

$$\text{Hence } z_1 = \frac{\left(x_1 - \frac{1}{2}\right) - \mu}{\sigma} = \frac{x_1 - \frac{1}{2} - np}{\sqrt{npq}}$$

$$\text{and } z_2 = \frac{\left(x_2 + \frac{1}{2}\right) - \mu}{\sigma} = \frac{x_2 + \frac{1}{2} - np}{\sqrt{npq}}$$

$\therefore$ The required probability = $\int_{z_1}^{z_2} \phi(z) dz$ and this can be evaluated using normal tables.

SOLVED EXAMPLES

Example 1 : Find the probability that out of 100 patients between 84 and 95 inclusive will survive a heart-operation given that the chances of survival is 0.9.

Solution : Here $p = 0.9$ and $n = 100$

$$\therefore q = 1 - p = 1 - 0.9 = 0.1$$

$$\text{Standard deviation} = \sqrt{npq} = \sqrt{90(0.1)} = \sqrt{9} = 3$$

$$\text{The required probability} = \sum_{r=84}^{95} {}^{100}C_r (0.9)^r (0.1)^{100-r}$$

We have to sum up a large number of terms of the binomial. To avoid it, we can replace the binomial distribution by a normal distribution.

Here $x_1 = 84$ and $x_2 = 95$. Hence while calculating z_1, z_2 corresponding to 84 and 95, we take $P(84 \leq x \leq 95) = P(83.5 < x < 95.8)$ and calculate z_1, z_2 correspondingly.

$$\text{Hence } z_1 = \frac{84 - (1/2) - 90}{3} = \frac{-13}{6} \text{ and } z_2 = \frac{95 + (1/2) - 90}{3} = \frac{11}{6}$$

$$\begin{aligned}
 \text{Hence the required probability} &= \int_{-13/6}^{11/6} \phi(z) dz \\
 &= P\left(-\frac{13}{6} \leq z \leq \frac{11}{6}\right) \\
 &= A\left(\frac{11}{6}\right) + A\left(\frac{-13}{6}\right) = A\left(\frac{11}{6}\right) + A\left(\frac{13}{6}\right) \\
 &= A(1.83) + A(2.17) \\
 &= 0.4664 + 0.4850 = 0.9514
 \end{aligned}$$

Example 2 : Eight coins are tossed together. Find the probability of getting 1 to 4 heads in a single toss.

Solution : Here $p = q = \frac{1}{2}$

$$\text{Mean, } \mu = np = 8 \times \frac{1}{2} = 4 \text{ and standard deviation, } \sigma = \sqrt{npq} = \sqrt{4 \times \frac{1}{2}} = \sqrt{2}$$

Even though $n = 8$ is not sufficiently large, we can approximate the symmetric binomial by a normal distribution.

Here $x_1 = 1$ and $x_2 = 4$

$$\text{Thus } z_1 = \frac{x_1 - \frac{1}{2} - \mu}{\sigma} = \frac{1 - \frac{1}{2} - 4}{\sqrt{2}} = \frac{-7}{2\sqrt{2}} = -2.47$$

$$\text{and } z_2 = \frac{x_2 + \frac{1}{2} - \mu}{\sigma} = \frac{4 + \frac{1}{2} - 4}{\sqrt{2}} = \frac{1}{2\sqrt{2}} = 0.35$$

$$\begin{aligned}
 \text{The required probability} &= \int_{x_1}^{x_2} \phi(z) dz = \int_{-2.47}^{0.35} \phi(z) dz \\
 &= P(-2.47 \leq z \leq 0.35) = A(0.35) + A(-2.47) \\
 &= A(0.35) + A(2.47) \\
 &= 0.4932 + 0.1368 = 0.63
 \end{aligned}$$

Example 3 : Find the probability that by guess-work a student can correctly answer 25 to 30 questions in a multiple-choice quiz consisting of 80 questions. Assume that in each question with four choices, only one choice is correct and student has no knowledge of the subject.

Solution : Here $p = \frac{1}{4}$. So $q = 1 - p = 1 - \frac{1}{4} = \frac{3}{4}$

$$\text{Mean, } \mu = np = 80 \left(\frac{1}{4}\right) = 20 \text{ and } \sigma = \sqrt{npq} = \sqrt{20 \times \frac{3}{4}} = \sqrt{15}, x_1 = 25 \text{ and } x_2 = 30$$

$$\text{Thus } z_1 = \frac{\left(x_1 - \frac{1}{2}\right) - \mu}{\sigma} = \frac{\left(25 - \frac{1}{2}\right) - 20}{3.87} = \frac{4.5}{3.87} = 1.16$$

$$\text{and } z_2 = \frac{\left(x_2 + \frac{1}{2}\right) - \mu}{\sigma} = \frac{\left(30 + \frac{1}{2}\right) - 20}{3.87} = \frac{10.5}{3.87} = 2.71$$

Hence the required probability = $P(25 \leq X \leq 30)$

$$= P(1.16 \leq z \leq 2.71) = |A(2.71) - A(1.16)|$$

$$= 0.4966 - 0.3770 = 0.1196$$

Example 4 : Find the probability of getting an even number on face 3 to 5 times in throwing 10 dice together. [JNTU (H) May 2012 (Set No. 2)]

Solution : p = Probability of getting an even number on face

$$= \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{3}{6} = \frac{1}{2}$$

$$\therefore q = 1 - p = \frac{1}{2}$$

Here $n = 10$

$$\text{Mean, } \mu = np = 10 \left(\frac{1}{2}\right) = 5$$

$$\text{and } \sigma = \sqrt{npq} = \sqrt{\frac{5}{2}} = 1.58$$

Also $x_1 = 3$ and $x_2 = 5$

$$\text{Hence } z_1 = \frac{\left(x_1 - \frac{1}{2}\right) - \mu}{\sigma} = \frac{(3 - 0.5) - 5}{1.58} = \frac{-2.5}{1.58} = -1.58$$

$$z_2 = \frac{\left(x_2 + \frac{1}{2}\right) - \mu}{\sigma} = \frac{(5 + 0.5) - 5}{1.58} = \frac{0.5}{1.58} = 0.32$$

$$\text{Hence the required probability} = \int_{x_1}^{x_2} \phi(z) dz = \int_{-1.58}^{0.32} \phi(z) dz$$

$$= P(-1.58 \leq z \leq 0.32)$$

$$= A(0.32) + A(1.58)$$

$$= 0.1256 + 0.4429 = 0.5685$$

7.17 FITTING OF NORMAL DISTRIBUTION

Consider the following frequency distribution

Class-interval (x)	Frequency (f)
$l_1 - u_1$	f_1
$l_2 - u_2$	f_2
$l_3 - u_3$	f_3
...	...
...	...
$l_i - u_i$	f_i
...	...
$l_n - u_n$	f_n

Then the normal curve (*i.e.*, Normal Distribution) can be fitted to it using

$$f(x) = \frac{1}{\sigma \cdot \sqrt{2\pi}} \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}}, -\infty < x < \infty$$

Working Procedure :

1. Calculate the mean μ and standard deviation σ from the given frequency distribution.
2. Compute the standard normal variate corresponding to the lower limits of each of the class interval. That is, find $z_i = \frac{x_i' - \mu}{\sigma}$, where x_i' is the lower limit of the i th class interval.
3. Compute the areas under the normal curve from 0 to z_i using normal tables.
4. Areas for the successive class intervals are obtained by taking the difference between the successive areas calculated in Step 3 (when z_i 's are of opposite sign, add the successive areas).
5. Expected or Theoretical normal frequencies are obtained by multiplying the areas in Step 4 by $N (= \sum f_i)$.

SOLVED EXAMPLES

Example 1 : Fit a normal distribution to the following data :

Class-interval	60-62	63-65	66-68	69-71	72-74
Frequency (f)	5	18	42	27	8

Solution : Here $N = \sum f_i = 100$, mean $= \mu = 67.45$ and standard deviation $= \sigma = 2.92$.
The calculations are arranged as follows :

S. No.	Class interval	Observed frequency (f)	True lower class limit x_i	Standard variate $z_i = (x_i - 67.45)/2.92$	Area from 0 to z_i	Area for each class (A)	Expected frequency = NA = 100A
1	60-62	5	59.5	-2.72	0.4967	0.0413	4.13
2	63-65	18	62.5	-1.70	0.4554	0.2068	20.68
			65.5	-0.67	0.2486	0.3892	38.92
3	66-68	42	68.5	0.36	0.1406	0.2771	27.71
4	69-71	27	71.5	1.39	0.4177	0.0743	7.43
5	72-74	8	74.5	2.41	0.4920		

Hence the expected frequencies corrected to nearest integers are 4, 21, 39, 28, 7 which agree with the observed frequencies.

REVIEW QUESTIONS

- (i) Define Normal Distribution.
(ii) Define Normal Distribution and derive Normal Distribution as a limiting form of Binomial distribution. [JNTU (K) Nov. 2011 (Set No. 1)]
- Write the uses of Normal Distribution
- List the properties of Normal Distribution.
- Write about the importance and applications of Normal distribution.
- (i) Derive Mean and Variance of normal distribution
(ii) Derive Mode and Median of normal distribution. [JNTU (A) Dec. 2017]

EXERCISE 7(C)

- The mean and standard deviation of a normal variate are 8 and 4 respectively, find
(i) $P(5 \leq x \leq 10)$ (ii) $P(x \geq 5)$ [JNTU 2005 S, (H) Nov. 2009 (Set No. 1)]
(iii) $P(10 \leq x \leq 15)$ (iv) $P(x \leq 15)$ [JNTU 2004 (Set No. 2)]
- If the mean and standard deviation of a normal distribution are 70 and 16, find $P(38 < x < 46)$. [JNTU 2005S (Set No. 4)]
- If X is normally distributed with mean 70 and standard deviation 16, find
(i) $P(38 \leq X \leq 46)$ (ii) $P(82 \leq X \leq 94)$ (iii) $P(62 \leq X \leq 86)$
- Find the probability that a random variable having standard normal distribution will take on a value between 0.87 and 1.28. [JNTU 2001]
- If X is a normally distributed with mean 30 and standard deviation 5, find $P(|X-30| > 5)$.
- Find the mean and standard deviation of a normal distribution in which 31% of items are under 45 and 8% are over 64. [Hint : Refer Example 4]
[JNTU 2003, (A) Dec 2009 (Set No. 2), (K) Nov. 2012, Dec. 2014 (Set No. 2)]
- (i) The mean height of 500 students is 151 cm. and the standard deviation is 15 cm. Assuming that the heights are normally distributed, find how many students heights lie between 120 and 155 cm.

- (ii) The heights of 1000 students are normally distributed with a mean of 174.5 cm and a standard deviation of 6.9 cm. Assuming that the heights are recorded to the nearest half-cm, how many of these students would you expect to have heights :
 (a) Less than 160.0 cms ? (b) Between 171.5 and 182.0 cms inclusive ?
 (c) Greater than or equal to 188.0 cms. [JNTU (A) June 2018]
8. The life of electronic tubes of a certain type may be assumed to be normally distributed with mean 155 hours and standard deviation 19 hours. Determine the probability that the life of a randomly chosen tube
 (i) is between 136 hours and 174 hours.
 (ii) less than 117 hours (iii) will be more than 395 hours.
9. In a test on 2000 electric bulbs, it was found that the life of a particular make, was normally distributed with an average life of 2040 hours and standard deviation of 60 hours. Find the number of bulbs likely to burn for
 (i) more than 2150 hours (ii) less than 1950 hours and
 (iii) more than 1920 hours and but less than 2160 hours.
10. Estimate the minimum marks a student must get in order to receive an 'A' grade if the top 10% of the students are awarded 'A' grade in an examination where the mean of the marks is 72 and standard deviation is 9.
11. When the mean of marks was 50% and standard deviation 5% then 60% of the students failed in an examination. Determine the 'grace' marks to be awarded in order to show that 70% of the students passed. Assume that the marks are normally distributed, [JNTU (A) May 2012 (Set No. 1)]
12. The mean and standard deviation of marks scored by a group of students in an examination are 15 and 1.5 respectively. If only 25% of the students are promoted, determine the minimum marks required for the promotion.
13. Find the probability of getting at least 55 heads when 100 coins are tossed using normal distribution.
14. Twelve coins are tossed together. Find the probability of getting 8, 9 or 10 heads in a single toss using normal distribution.
15. Ten cards are drawn from a deck of 52 cards. Find the probability of getting 2 to 5 diamonds using normal distribution.
16. In a referendum 60% of voters voted in favour. A random sample of 200 voters was selected. Find the probability by using normal distribution that in the sample
 (i) more than 130 voted in favour ?
 (ii) between 105 and 130 inclusive voted in favour ?
 (iii) 120 voted in favour ?
17. Determine the probability of getting at least 60 heads when 100 coins are tossed.
18. The weekly wages of 1000 workers are normally distributed around a mean of Rs. 70 and S.D. of Rs. 5/-. Estimate the number of workers whose weekly wages will be
 (i) between Rs. 70 and Rs. 72
 (ii) between Rs. 69 and Rs. 72. [JNTU 2007S (Set No.1)]

19. A sample of 100 dry battery cells tested to find the length of life produced the following results : $\bar{x} = 12$ hours, $\sigma = 2$ hours

Assuming the data to be normally distributed, what percentage of battery cells are expected to have life

- (i) More than 15 hours (ii) Less than 6 hours
(iii) Between 10 and 15 hours ?

Fit a normal distribution to the following data :

x	2	4	6	8	10
f	1	4	6	4	1

21. Fit a normal curve to the following distribution :

Class	5 - 9	10 - 14	15 - 19	20 - 24	25 - 29	30 - 34	35 - 39
Frequency	1	10	37	36	13	2	1

22. Fit a normal distribution to the following data :

x	2	4	6	8	10
f	1	4	6	4	1

ANSWERS

1. (i) 0.465 (ii) 0.7734 (iii) 0.6514 (iv) 0.0401
2. 0.044
3. (i) 0.044 (ii) 0.1598 (iii) 0.5328
4. 0.0919
5. 0.3174
6. $\mu = 50, \sigma = 10$
7. 294
8. (i) 0.6826 (ii) 0.0228 (iii) 0
9. (i) 67 (ii) 184 (iii) 1909
10. 84
11. 3.85
12. 28
13. 0.1841
14. 0.1887
15. 0.7530
16. (i) 0.0648 (ii) 0.8964 (iii) 0.0575
17. 0.02872
18. 155, 235
20. 1, 4, 6, 4, 1
21. 2, 1, 3, 36, 16, 3, 0
22. 1, 4, 6, 4, 1

3.3 CHEBYSHEV'S THEOREM (LAW OF LARGE NUMBERS)

[1999/S, 2001/S, Jan 2003, Dec. 2002 (Set-4)]

Let X be a random variable with mean μ and standard deviation σ then

$$P(|x - \mu| \geq k\sigma) \leq \frac{1}{k^2}$$

Proof:

Let $f(x)$ be the probability mass function of a random variable having mean μ and variance

σ^2 .

$$\text{Now } \sigma^2 = \sum_x (x - \mu)^2 f(x) \quad \dots\dots\dots (1)$$

Let R_1 be the region in which $x \leq \mu - k\sigma$, R_2 be the region in which $\mu - k\sigma < x < \mu + k\sigma$ and R_3 be the region in which $x \geq \mu + k\sigma$.

$$\therefore \sigma^2 = \sum_{R_1} (x - \mu)^2 f(x) + \sum_{R_2} (x - \mu)^2 f(x) + \sum_{R_3} (x - \mu)^2 f(x) \quad \dots\dots\dots (2)$$

Since $(x - \mu)^2 f(x)$ is always non-negative, hence the above summation over R_2 is non-negative.

Hence from (2)

$$\sigma^2 \geq \sum_{R_1} (x - \mu)^2 f(x) + \sum_{R_3} (x - \mu)^2 f(x) \quad \dots\dots\dots (3)$$

$$\begin{aligned} \text{In } R_1 \quad x &\leq \mu - k\sigma \\ \Rightarrow \quad x - \mu &\leq -k\sigma \end{aligned} \quad \dots\dots\dots (4)$$

$$\begin{aligned} \text{In } R_3 \quad x &\geq \mu + k\sigma \\ \Rightarrow \quad x - \mu &\geq k\sigma \end{aligned} \quad \dots\dots\dots (5)$$

Hence from (4) and (5)

$$|x - \mu| \geq k\sigma$$

Hence in both R_1 & R_3 , $(x - \mu)^2 \geq k^2 \sigma^2$

$\therefore$ From (3)

$$\sigma^2 \geq \sum_{R_1} k^2 \sigma^2 f(x) + \sum_{R_3} k^2 \sigma^2 f(x) \quad \dots\dots\dots (6)$$

$$\text{i.e., } \sigma^2 \geq k^2 \sigma^2 \left[\sum_{R_1} f(x) + \sum_{R_3} f(x) \right]$$

$$\Rightarrow \frac{1}{k^2} \geq \sum_{R_1} f(x) + \sum_{R_3} f(x) \quad \dots\dots\dots (7)$$

Now $\sum_{R_1} f(x) + \sum_{R_3} f(x)$ represents the probability assigned to the region $R_1 \cup R_3$.

$$\therefore \sum_{R_1} f(x) + \sum_{R_3} f(x) = P[|x - \mu| \geq k\sigma] \quad \dots\dots\dots (8)$$

$\therefore$ From (7) and (8)

$$\frac{1}{k^2} \geq P[|X - \mu| \geq k\sigma] \Rightarrow P[|X - \mu| \geq k\sigma] \leq \frac{1}{k^2}$$

Corollary :

From the above theorem

$$P[|X - \mu| < k\sigma] \geq 1 - \frac{1}{k^2}$$

SOLVED PROBLEMS

PROBLEM 1

The number of customers who visit a car dealer show room on a certain day is a random variable with mean 18 and standard deviation 2.5. With what probability can it be asserted that there will be between 8 and 28 customers.

Solution :

$$\mu = 18, \quad \sigma = 2.5$$

$$k\sigma = |x - \mu|$$

$$\therefore k = \frac{|x - \mu|}{\sigma}$$

$$\text{When } x = 28, \quad k = \frac{|28 - 18|}{2.5} = 4$$

$$\text{When } x = 8, \quad k = \frac{|8 - 18|}{2.5} = 4$$

$\therefore$ From the corollary

$$P[|X - \mu| < k\sigma] \geq 1 - \frac{1}{k^2}$$

$$\text{i.e., } P[|X - \mu| < 4\sigma] \geq 1 - \frac{1}{16}$$

$$\Rightarrow P[|X - \mu| < 4\sigma] \geq \frac{15}{16}$$

PROBLEM 2

If X is the number appearing on a die when it is thrown, show that the Chebyshev's theorem gives

$$P[|X - \mu| > 2.5] < 0.47$$

while the actual probability is zero.

Solution :

Let X denote the number appearing on the die. The probability distribution of X is

$X = x$	1	2	3	4	5	6
$P(X = x)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

$$\mu = E(X) = 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + 3 \cdot \frac{1}{6} + 4 \cdot \frac{1}{6} + 5 \cdot \frac{1}{6} + 6 \cdot \frac{1}{6}$$

$$= \frac{1}{6} (1 + 2 + 3 + 4 + 5 + 6) = \frac{7}{2}$$

$$E(X^2) = 1^2 \cdot \frac{1}{6} + 2^2 \cdot \frac{1}{6} + 3^2 \cdot \frac{1}{6} + 4^2 \cdot \frac{1}{6} + 5^2 \cdot \frac{1}{6} + 6^2 \cdot \frac{1}{6}$$

$$= \frac{1}{6} (1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2) = \frac{91}{6}$$

$$\sigma^2 = V(X) = E(X^2) - [E(X)]^2 = \frac{91}{6} - \left(\frac{7}{2}\right)^2 = \frac{35}{12} = 2.9167$$

$$\therefore \sigma = 1.708$$

$$\text{Now } P[|X - \mu| \geq k\sigma] < \frac{1}{k^2}$$

$$\text{Let } k\sigma = 2.5$$

$$\therefore k = \frac{2.5}{\sigma} = \frac{2.5}{1.708} = 1.464$$

$$\therefore P[|X - \mu| \geq 2.5] < \frac{1}{(1.464)^2}$$

$$\therefore P[|X - 3.5| \geq 2.5] < 0.47$$

But $P[|X - 3.5| \geq 2.5]$ is the probability that the random variable X lies outside the limits $(3.2 - 2.5, 3.5 + 2.5)$ i.e., $(1, 6)$ which is zero for a die.

PROBLEM 3

Use Chebyshev's inequality to find a lower bound on $P(-4 < X < 20)$ where the random variable X has a mean $\mu = 8$ and variance $\sigma^2 = 9$.

Solution :

Chebyshev's inequality gives

$$P[|X - \mu| < k\sigma] \geq 1 - \frac{1}{k^2} \quad \text{or}$$

$$P[\mu - k\sigma < X < \mu + k\sigma] \geq 1 - \frac{1}{k^2}$$

we wish to find k

Let $\mu - k\sigma = -4$ and $\mu + k\sigma = 20$

Given $\mu = 8$ and $\sigma^2 = \sqrt{9} = 3$

k satisfies either

$$8 - 3k = -4 \text{ or } 8 + 3k = 20$$

Hence $k = 4$

$$\text{Then } P[\mu - k\sigma < X < \mu + k\sigma] \geq 1 - \frac{1}{k^2}$$

$$\Rightarrow P[-4 < X < 20] \geq 1 - \frac{1}{(4)^2} = 1 - \frac{1}{16} = \frac{15}{16}$$

Thus a lower bound on $P[-4 < X < 20]$ is $\frac{15}{16}$

PROBLEM 4

Find a lower bound on $P[-3 < X < 3]$ where $\mu = E(X) = 0$ and variance $= \sigma^2 = 1$.

Solution :

From Chebyshev's inequality gives

$$P[|X - \mu| < k\sigma] \geq 1 - \frac{1}{k^2} \quad \text{or}$$

$$P[\mu - k\sigma < X < \mu + k\sigma] \geq 1 - \frac{1}{k^2}$$

we wish to find k

Let $\mu - k\sigma = -3$

we know that $\mu = 0$ and $\sigma = 1$

$$\Rightarrow 0 - k(1) = -3$$

Therefore $k = 3$

$$\text{Thus } P[0 - 3 < X < 0 + 3] \geq 1 - \frac{1}{3^2}$$

$$P[-3 < X < 3] \geq 1 - \frac{1}{9} = \frac{8}{9}$$

The lower bound is $\frac{8}{9}$

Suppose that X assumes the values 1 and -1, each with probability 0.5. Find and compare the lower bound on $P[-1 < X < 1]$ given by Chebyshev's inequality and the actual probability that $-1 < X < 1$.

Solution :

Chebyshev's inequality gives

$$P[\mu - k\sigma < X < \mu + k\sigma] \geq 1 - \frac{1}{k^2}$$

for this random variables

$$\begin{aligned}\mu &= E(X) = P(X=1) \cdot (1) + P(X=-1) \cdot (-1) \\ &= 0.5 - 0.5 = 0\end{aligned}$$

$$\begin{aligned}\text{and variance } \sigma^2 &= E(X^2) - [E(x)]^2 \\ &= E(X^2)\end{aligned}$$

$$\begin{aligned}\therefore E(X^2) &= P(X=1)(1)^2 + P(X=-1)(-1)^2 \\ &= 0.5 + 0.5 = 1\end{aligned}$$

$$\text{Thus variance} = \sigma^2 = 1 \Rightarrow \sigma = 1$$

$$\text{Now } P[0 - k < X < 0 + k] \geq 1 - \frac{1}{k^2}$$

$$\text{If } k = 1, \text{ then } P[-1 < X < 1] \geq 1 - \frac{1}{1} = 0$$

$$\text{Thus } P[-1 < X < 1] \geq 0$$

The actual probability that X is between 1 and -1 is found by

$$P[-1 \leq X \leq 1] = P[X=1] + P[-1 < X < 1] + P[X=-1]$$

But $P[-1 \leq X \leq 1] = 1$ because -1 and 1 are the values that X assumes with positive probability that is.

$$1 = 0.5 + P[-1 < X < 1] + 0.5 \text{ or } P[-1 < X < 1] = 0$$

Thus, in this case the lower bound on this probability equals the true probability when the distribution of X is known.

PROBLEM 6

Two unbiased dice are thrown. If X is the sum of the numbers showing up. Prove that

$$P(|X - 7| \geq 3) \leq \frac{35}{54}$$

Compare this with the actual probability.

[JNTU 2000]

Solution :

The probability distribution of the random variable X (the sum of the numbers on the two dice) is as given below:

X	Favourable Cases (Distinct)	Probability
2	(1, 1)	$\frac{1}{36}$
3	(1, 2), (2, 1)	$\frac{2}{36}$
4	(1, 3), (3, 1), (2, 2)	$\frac{3}{36}$
5	(1, 4), (4, 1), (2, 3), (3, 2)	$\frac{4}{36}$
6	(1, 5), (5, 1), (2, 4), (4, 2), (3, 3)	$\frac{5}{36}$
7	(1, 6), (6, 1), (2, 5), (5, 2), (3, 4), (4, 3)	$\frac{6}{36}$
8	(2, 6), (6, 2), (3, 5), (5, 3), (4, 4)	$\frac{5}{36}$
9	(3, 6), (6, 3), (4, 5), (5, 4)	$\frac{4}{36}$
10	(4, 6), (6, 4), (5, 5)	$\frac{3}{36}$
11	(5, 6), (6, 5)	$\frac{2}{36}$
12	(6, 6)	$\frac{1}{36}$

$$E(X) = \sum_x xp(x)$$

$$= \frac{1}{36} (2 + 6 + 12 + 20 + 30 + 42 + 40 + 36 + 30 + 22 + 12)$$

$$= \frac{1}{36} (252) = 7$$

$$E(X^2) = \sum_x x^2 P(x)$$

$$= \frac{1}{36} [4 + 18 + 48 + 100 + 180 + 294 + 320 + 324 + 300 + 242 + 144]$$

$$= \frac{1}{36} (1974) = \frac{329}{6}$$

Variance $\sigma^2 = V(X) = E(X^2) - [E(X)]^2 = \frac{329}{6} - (7)^2 = \frac{35}{6}$

By Chebyshev's inequality for $k > 0$

We have $P(|X - \mu| \geq k) \leq \frac{V(X)}{k^2}$

Taking $k = 3$

$$\Rightarrow P(|X - 7| \geq 3) \leq \frac{35/6}{9} = \frac{35}{54}$$

Actual Probability

$$P(|X - 7| \geq 3) = 1 - P(|X - 7| < 3)$$

$$= 1 - P(4 < X < 10)$$

$$= 1 - [P(X=5) + P(X=6) + P(X=7) + P(X=8) + P(X=9)]$$

$$= 1 - \frac{1}{36} [4 + 5 + 6 + 5 + 4] = 1 - \frac{24}{36} = \frac{1}{3}$$

PROBLEM 7

For geometric distribution $P(x) = 2^{-x}$; $x = 1, 2, \dots$ prove that Chebyshev's inequality gives

$$P(|X - 2| \leq 2) > \frac{1}{2}$$

while the actual probability is $\frac{15}{16}$

Solution :

$$E(X) = \sum_{x=1}^{\infty} \frac{x}{2^x} = \frac{1}{2} + \frac{2}{2^2} + \frac{3}{2^3} + \dots$$

$$= \frac{1}{2} \left(1 + \frac{2}{2} + \frac{3}{4} + \frac{4}{8} + \dots \right) = \frac{1}{2} \left(1 - \frac{1}{2} \right)^{-2} = 2$$

$$E(X^2) = \sum_{x=1}^{\infty} \frac{x^2}{2^x} = \frac{1}{2^2} + \frac{4}{2^3} + \frac{9}{2^4} + \dots$$

$$= \frac{1}{4} \left[1 + \frac{4}{2} + \frac{9}{4} + \dots \right] = \frac{1}{4} \left(1 + \frac{1}{2} \right) \left(1 - \frac{1}{2} \right)^{-3} = 6$$

$$\therefore \text{Variance} = \sigma^2 = V(X) = E(X^2) - [E(X)]^2 = 6 - 4 = 2$$

From Chebyshev's inequality we have

$$P[|X - \mu| \leq k\sigma] > 1 - \frac{1}{k^2}$$

with $k = \sqrt{2}$, we get

$$P[|X - 2| \leq \sqrt{2} \cdot \sqrt{2}] > 1 - \frac{1}{2} = \frac{1}{2}$$

$$\Rightarrow P[|X - 2| \leq 2] > \frac{1}{2}$$

and the actual probability is given by

$$P[|X - 2| \leq 2] = P[0 \leq X \leq 4] = P[X = 1, 2, 3 \text{ or } 4]$$

$$= \frac{1}{2} + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^3 + \left(\frac{1}{2}\right)^4 = \frac{15}{16}$$

PROBLEM 8

Given that the discrete random variable X has density function $f(x)$ given by $f(-1) = \frac{1}{8}, f(0) = \frac{6}{8}, f(1) = \frac{1}{8}$ use Chebyshev's inequality to find the upper bound when $k = 2$.

Solution :

Chebyshev's inequality states that

$$P[|X - \mu| \geq k\sigma] \leq \frac{1}{k^2}$$

In order to use the inequality, we need to know the mean and variance of X .

$$\mu = E(X) = \sum_x x f(x)$$

$$= (-1) \left(\frac{1}{8}\right) + (0) \left(\frac{6}{8}\right) + (1) \left(\frac{1}{8}\right) = -\frac{1}{8} + \frac{1}{8} = 0$$

$$\text{Variance } \sigma^2 = E[(X - \mu)^2] = E[(X - 0)^2] = E(X^2)$$

$$\sum_x x^2 f(x) = (-1)^2 \left(\frac{1}{8}\right) + 0 + (1)^2 \left(\frac{1}{8}\right) = \frac{1}{8} + \frac{1}{8} = \frac{1}{4}$$

$$\text{when } k = 2 \quad P[|X - \mu| \geq k\sigma] = P[|X| \geq 2 \sqrt{(1/4)}] = P[|X| \geq 1] \leq \frac{1}{2^2} = \frac{1}{4}$$

Thus the upper bound when $k = 2$ is $\frac{1}{4}$

PROBLEM 9

DISTRIBUTIONS

UNIT III 3-37

A symmetric die is thrown 600 times. Find the lower bound for the probability of getting 80 to 120 sixes.

Solution :

Let X be total number of successes.

Here $n = 600$

Probability of getting six = $P = \frac{1}{6}$ and probability of not getting six = $q = 1 - \frac{1}{6} = \frac{5}{6}$

$$\mu = E(X) = np = 600 \times \frac{1}{6} = 100$$

$$\sigma^2 = \text{variance} = npq = 600 \times \frac{1}{6} \times \frac{5}{6} = \frac{500}{6}$$

$$\therefore \sigma = \sqrt{\frac{500}{6}}$$

From Chebyshev's inequality we get

$$P[|X - \mu| < k\sigma] \geq 1 - \frac{1}{k^2} \Rightarrow P\left[|X - 100| < k\sqrt{\frac{500}{6}}\right] \geq 1 - \frac{1}{k^2}$$

$$P\left[|X - 100| < k\sqrt{\frac{500}{6}}\right] \geq 1 - \frac{1}{k^2}$$

$$\Rightarrow P\left[100 - k\sqrt{\frac{500}{6}} < X < 100 + k\sqrt{\frac{500}{6}}\right] \geq 1 - \frac{1}{k^2}$$

Taking $k = \frac{-20}{\sqrt{500/6}}$, we get

$$\therefore P(80 \leq X \leq 120) \geq 1 - \frac{1}{400 \times \frac{6}{500}} = \frac{19}{24}$$

PROBLEM 10

500 articles were selected at random out of a batch containing 10000 articles and 30 were found to be defective. How many defective articles would you reasonably expect to have in the whole batch?

[JNTU, April 2003]

Solution :

$$\text{Here } p = \frac{30}{400} = \frac{3}{50}$$

$$N = 10000$$

$$\therefore E(X) = Np = 10000 \times \frac{3}{50} = 600$$

PROBABILITY AND STATISTICS

PROBLEM 11

X is a random variable such that $E(X) = 3$ and $E(X^2) = 13$. Determine a lower bound for $P(-2 < x < 8)$, using Chebyshev's inequality.

[JNTU, Jan. 2003]

Solution:

$$E(X) = 3, E(X^2) = 13$$

$$\sigma^2 = E(X^2) - [E(X)]^2 = 13 - (3)^2 = 13 - 9 = 4$$

$$\therefore \sigma = 2$$

By Chebyshev's inequality

$$P[|X - \mu| < \sigma K] \geq 1 - \frac{1}{K^2}$$

$$\therefore P[|X - 3| < 2K] \geq 1 - \frac{1}{K^2}$$

$$\therefore P(3 - 2K < X < 3 + 2K) \geq 1 - \frac{1}{K^2}$$

Taking $K = 2.5$, we get

$$P(-2 < X < 8) \geq \frac{21}{25}$$

$$\therefore \text{Lower bound is } \frac{21}{25}$$

EXERCISE

1. A random variable X has density function given by

$$f(x) = \begin{cases} 2e^{-2x} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

- (a) Find $P[|X - \mu| > 1]$

- (b) Use Chebyshev's inequality to obtain an upper bound on $P(|X - \mu| > 1)$ and compare with the result in (a).

Ans. (a) $e^{-3} = 0.04979$

(b) 0.25

2. Prove Chebyshev's inequality for a discrete variable X

3. Let $X_1, X_2, \dots, X_n$ be n independent random variables each having density function

$$f(x) = \begin{cases} \frac{1}{2} & -1 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\text{If } S_n = X_1 + X_2 + \dots + X_n$$

Show that

$$P\left[\left|\frac{S_n}{n} - \mu\right| < \epsilon\right] = 1$$

4. A random variable X has mean 3 and variance 2. Use Chebyshev's inequality to obtain an upper bound for.

(a) $P[|X - 3| \geq 2]$

(b) $P[|X - \mu| \geq 1]$

Ans. (a) 1 (b) $\frac{1}{4}$

5. A random variable X has the density function

$$f(x) = \frac{1}{2} e^{-|x|}, \quad -\infty < x < \infty$$

(a) Find $P[|X - \mu| > 2]$

- (b) Use Chebyshev's inequality to obtain an upper bound on $P[|X - \mu| > 2]$ and compare with the result in (a)

Ans. (a) e^{-2} (b) 0.5